EXP.NO:5
14/8/26

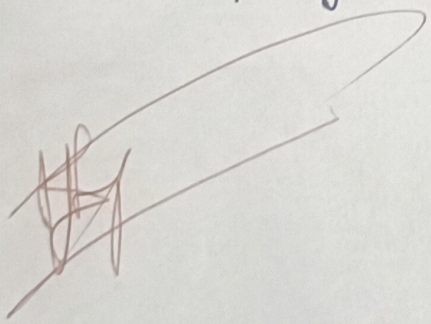
DEVELOP A CUSTOMIZED PING COMMAND TO TEST THE SERVER CONNECTIVITY

AIM:

To develop a customized ping command in python that checks the connectivity between a client and server by sending ICMP echo request packets and displaying the response time and status

STEPS

1. Install Python and vs code
2. Create a python file named ping.py
3. Import the required modules
4. Accept the hostname or IP address
5. Detect the operating system
6. Use ping -n 4 on window or ping -c 4 on linux / mac OS
7. Execute the ping command
8. Display ping output
9. Display Reachable or unreachable



RESULT:

Thus a ping command in python that checks connectivity between client and server and displayed.

```
PS C:\Users\REC.DESKTOP-ELL365E\echo> python ping.py
================================================================

Customized Ping Utility

================================================================

Enter Website or IP Address: google.com

Ping Results


Pinging google.com [142.251.221.142] with 32 bytes of data:
Reply from 142.251.221.142: bytes=32 time=5ms TTL=120
Reply from 142.251.221.142: bytes=32 time=5ms TTL=120
Reply from 142.251.221.142: bytes=32 time=20ms TTL=120
Reply from 142.251.221.142: bytes=32 time=41ms TTL=120

Ping statistics for 142.251.221.142:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 41ms, Average = 17ms
```

ping.py

```
import subprocess
import platform

print("=" * 50)
print("Customized Ping Utility")
print("=" * 50)

host = input("Enter Website or IP Address: ")

system = platform.system()

if system == "Windows":
    command = ["ping", "-n", "4", host]
else:
    command = ["ping", "-c", "4", host]

try:
    result = subprocess.run(command, capture_output=True, text=True

    print("\nPing Results\n")
    print(result.stdout)

    if result.returncode == 0:
        print("Server Status : Reachable")
    else:
        print("Server Status : Unreachable")

except Exception as e:
    print("Error:", e)
```